

tf.compat.v1.math.sign Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/sign

Returns an element-wise indication of the sign of a number.

Function Signatures

```
tf.compat.v1.math.sign( x, name=None )  
y = sign(x) = -1 if x < 0; 0 if x == 0; 1 if x > 0  
y = sign(x) = x / |x| if x != 0, otherwise y = 0  
tf.math.sign([0., 2., -3.])
```

Code Examples

```
tf.compat.v1.math.sign(  
    x, name=None  
)
```

```
tf.compat.v1.math.sign(  
    x, name=None  
)
```

```
# real number  
tf.math.sign([0., 2., -3.])
```

```
# real number
```

```
tf.math.sign([0., 2., -3.])
```

```
numpy=array([ 0.,  1., -1.], dtype=float32)>
```

```
# complex number  
tf.math.sign([1 + 1j, 0 + 0j])
```

Documentation

Returns an element-wise indication of the sign of a number.

$y = \text{sign}(x) = -1$ if $x < 0$; 0 if $x == 0$; 1 if $x > 0$.

For complex numbers, $y = \text{sign}(x) = x / |x|$ if $x \neq 0$, otherwise $y = 0$.

Example usage:

Returns

If x is a `SparseTensor`, returns `SparseTensor(x.indices, tf.math.sign(x.values, ...), x.dense_shape)`.

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`SparseTensor(x.indices, tf.math.sign(x.values, ...), x.dense_shape)`